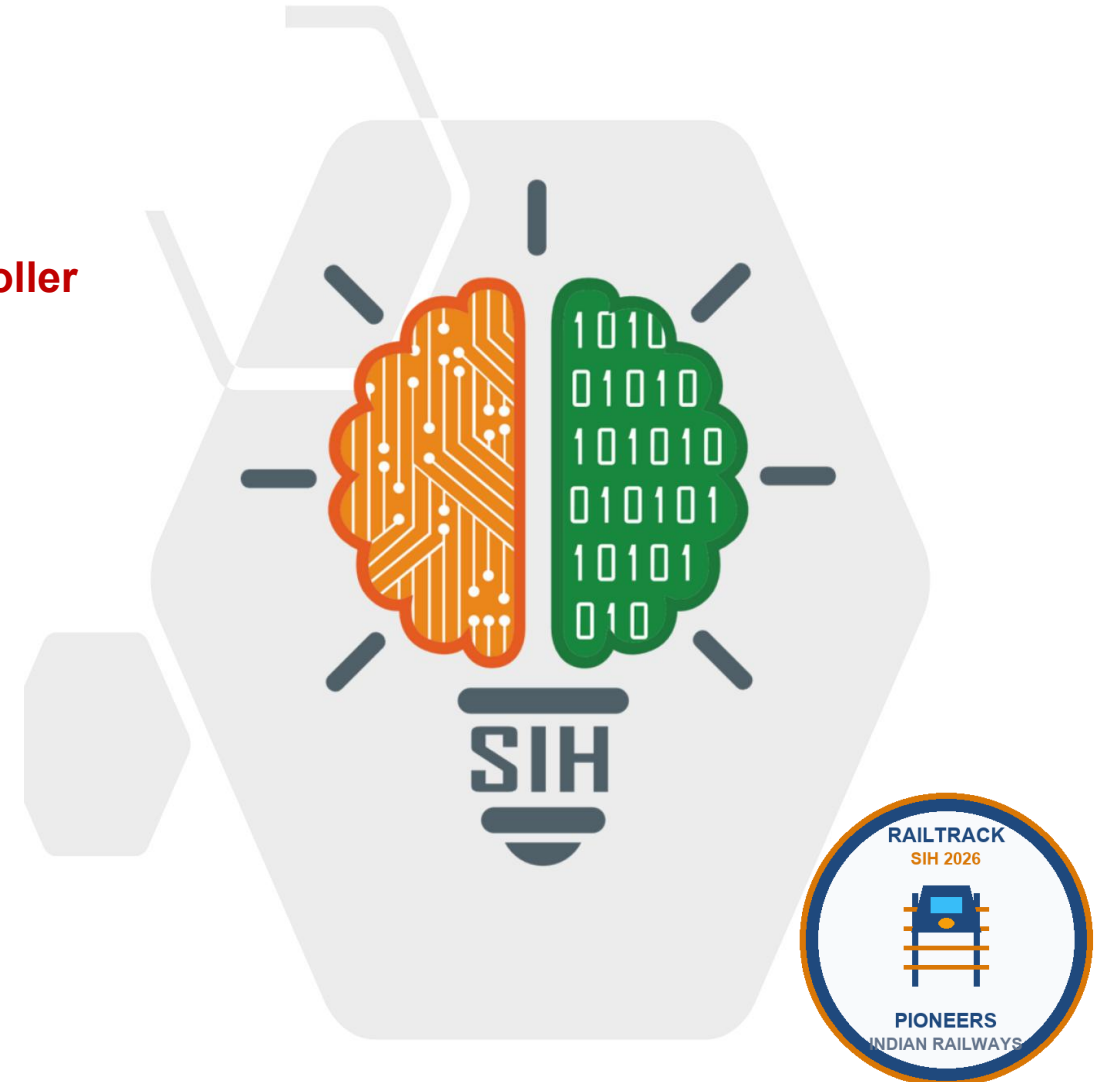


SMART INDIA HACKATHON 2026

TITLE PAGE



- Problem Statement ID – **SIH26028**
- Problem Statement Title- **AI-Driven Dynamic Section Controller & Track Maintenance Block Scheduling System for Indian Railways**
- PS Category- **Software**
- Theme- Transportation & Logistics / Smart Railways
- Team ID- SIH-26028-TEAM
- Team Name (Registered on portal) : **RailTrack Pioneers**





AI-Driven Section Controller & Maintenance Block Planner

❖ Proposed Solution



We Propose:

- **Autonomous Block Scheduling:** Constraint Programming (Google OR-Tools CP-SAT) dynamically computes conflict-free 2-4 hour track possession windows without cancelling express trains.
- **Kinematic Conflict Gating:** Continuous GPS/NavIC tracking with mathematical 5-minute safety headway enforcement between trains and heavy track machines.
- **Multi-Model ETA Predictor:** SLSQP + XGBoost meta-ensemble predicting dynamic section arrival times with sub-2-minute accuracy.

Uniqueness:

- **Closed-Loop Dual Control:** Section Controller Gantt workbench + passenger transparency portal.
- **Dynamic IMD Weather Guardrail:** Automatically expands safety buffers (+5m/+10m) upon adverse rain/fog alerts.

• Mathematical G&SR Compliance:

Enforces Indian Railways Block Working Rules and automatic signaling blocks.

• Sub-1.2s Optimization Speed:

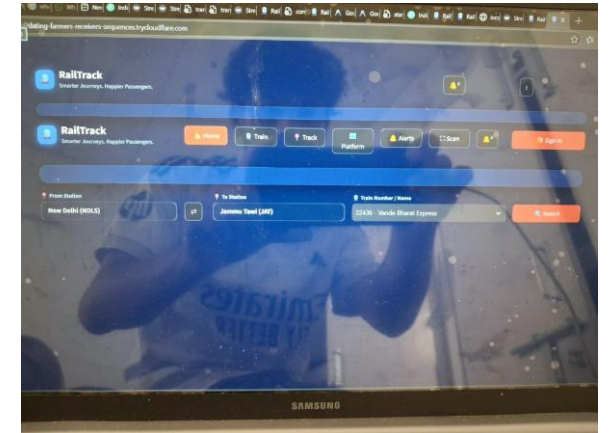
Evaluates 40+ track parameters simultaneously on standard laptops.

• Local SQLite WAL Continuity:

Ensures 100% offline continuity during fiber or WAN connectivity failures.

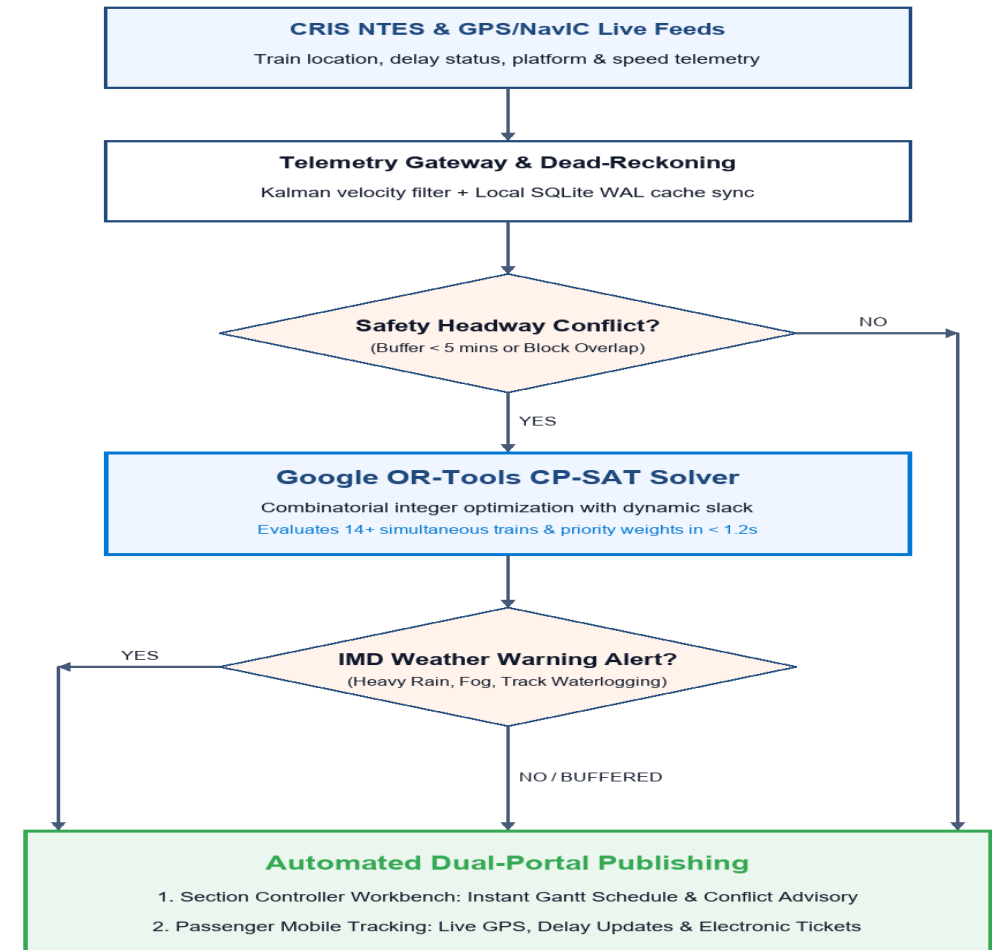
• Multi-Corridor Scalability:

Tested on high-density New Delhi – Ludhiana – Jammu Tawi section.



Core Technology Stack & Architecture

- **Optimization Engine:** Google OR-Tools 9.10 (CP-SAT Solver) with integer linear constraints and dynamic slack.
- **Backend Services:** FastAPI (Python 3.11), Uvicorn ASGI, REST APIs, WebSockets for live telemetry broadcast.
- **Machine Learning:** Dynamic ETA Predictor: Meta-Ensemble (SLSQP, XGBoost, Random Forest) trained on IR timetable data.
- **Data Ingestion:** CRIS NTES connector, Open-Meteo REST API, and IMD Weather Telemetry client with fallback cache.
- **Database & Persistence:** Supabase Cloud PostgreSQL with resilient local SQLite WAL offline auto-sync.
- **Frontend & Maps:** Streamlit responsive mobile/desktop dashboard, Plotly interactive OpenStreetMap GIS tracking.



Challenges & Solutions

Operational Feasibility: Controller workflow requires zero retraining.

Fits standard Control Office Application (COA) layout.

Replaces chaotic phone calls with single-click advisory.

Supports dynamic maintenance window adjustments.

Technical Feasibility: FastAPI + Google OR-Tools + Scikit-Learn.

Sub-1.2s optimization solves full section schedules.

100% offline continuity with SQLite WAL database.

Seamless integration with ISRO NavIC GPS telemetry.

Economic Feasibility: Zero trackside hardware procurement cost.

Uses existing locomotive GPS & CRIS NTES feeds.

Recovers ₹100s of crores in avoided idle fuel.

Maximizes return on high-speed rail corridors.

- GPS drift & telemetry packet loss in remote railway terrains

- Sudden extreme monsoon rains and severe winter fog

- Congested mixed traffic: high-speed Vande Bharat vs freight

- High track worker safety risks during emergency possessions

- Kinematic dead-reckoning filter with local SQLite WAL sync

- Automated IMD weather guardrail injecting dynamic buffers (+5m/+10m)

- Priority-weighted CP-SAT solver protects high-priority express slots

- Cryptographic token commit gating before track section lock

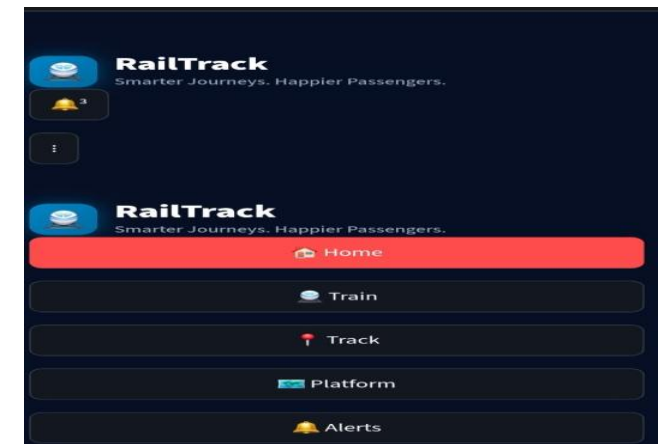
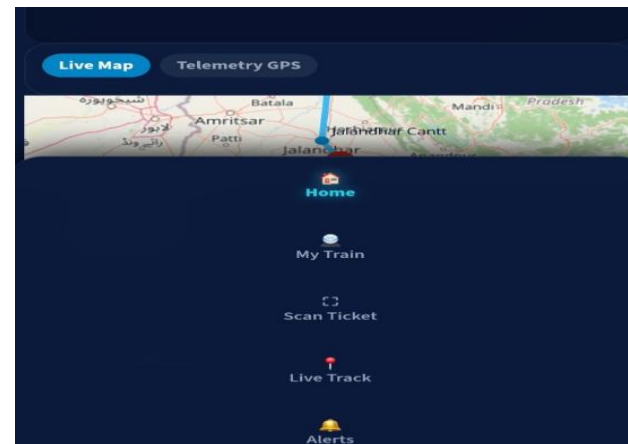
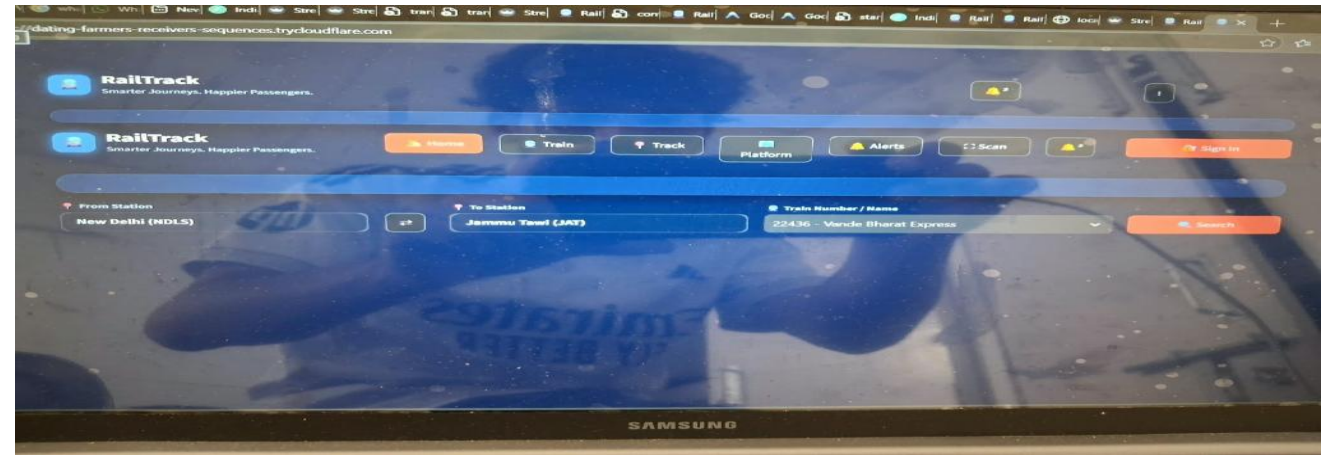
Potential Impact (Dynamic ETA Precision):

- **Sub-2-Minute ETA Accuracy:** Meta-ensemble (SLSQP + XGBoost) predicts dynamic section arrival & clearance times with sub-2-min precision.
- **Zero Cascading Delays:** Real-time dynamic ETA updates eliminate false loop-line siding halts and stop knock-on delay propagation.
- **Conflict-Free Block Windows:** High-confidence dynamic ETAs enable seamless insertion of 2-4 hr maintenance blocks without disrupting express trains.
- **Live Passenger Transparency:** Replaces static timetable drift with live, weather-aware dynamic ETA countdowns on passenger portals.

Benefits (Dynamic ETA Powered):

- **Economic: 35% Idling Reduction:** Dynamic ETA dispatching minimizes loop-line waiting, boosting section line capacity to 92% and saving ₹100s of crores in diesel and electric traction energy.
- **Safety & Social: Guaranteed Headway Gating:** Dynamic ETA calculations enforce a strict 5-min safety headway buffer before track possession, protecting maintenance workers and eliminating collision risks.
- **Environmental: Lower Carbon Emissions:** Optimized speed trajectories and reduced outer-signal idling drastically curb locomotive diesel burn and brake-shoe particulate emissions.

PROTOTYPE IMAGES (DYNAMIC ETA WORKBENCH & TRACKING)





RESEARCH AND REFERENCES



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